MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY TEST – 03** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Short Answer Type-I question of 2 marks.
4. **Section C** comprises 3 Short Answer Type-II questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (5 × 1 = 5 Marks)

Q1. If $\cos\theta = \frac{2}{3}$, then $2\sec^2\theta + 2\tan^2\theta - 7$ is equal to: [1]

(A) 0 (B) 1 (C) 3 (D) 4

Q2. If $\cot\theta = \frac{7}{8}$, then the value of $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$ is: [1]

(A) $\frac{49}{64}$ (B) $\frac{64}{49}$
(C) $\frac{7}{8}$ (D) $\frac{8}{7}$

Q3. In ΔPQR, $\angle Q = 90^\circ$, $PQ = 12$ cm, and $QR = 5$ cm. The value of $\tan P - \cot R$ is: [1]

(A) 0 (B) $\frac{5}{12}$
(C) $\frac{12}{5}$ (D) $\frac{119}{60}$

Q4. Which of the following can extbfnever be the value of $\sec\theta$ for any acute angle θ? [1]

(A) $\frac{1}{2}$ (B) 2 (C) $\frac{5}{4}$ (D) 3

Q5. **Assertion (A):** $\sin(A+B) = \sin A + \sin B$ for all angles A and B.
Reason (R): If $A = 30^\circ$ and $B = 60^\circ$, $\sin(A+B) = \sin 90^\circ = 1$, whereas $\sin 30^\circ + \sin 60^\circ = \frac{1+\sqrt{3}}{2} \neq 1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type-I (1 × 2 = 2 Marks)

Q6. Evaluate: $\frac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$. [2]

SECTION C: Short Answer Type-II (3 × 3 = 9 Marks)

Q7. Evaluate: $\frac{\sin 30^\circ + \tan 45^\circ - \csc 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$. [3]

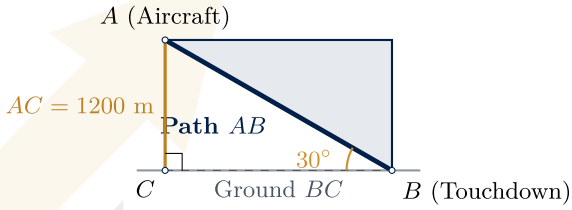
Q8. Prove that: $\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$. [3]

Q9. Prove that: $(\sin A + \csc A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$. [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Precision Instrument Flight Path Runway Glide Slope

During final approach, an aircraft follows a calibrated instrument glide slope. At a particular point in its descent, the aircraft A is at an altitude of 1200 m directly above ground point C. The flight path AB to the touchdown zone B is inclined at an angle of $\theta = 30^\circ$ to the horizontal ground line BC.



Based on the above geometric model, answer the following questions:

(i) State the trigonometric ratio relating altitude AC and direct flight distance AB. [1]

(ii) Calculate the direct straight-line distance AB remaining between the aircraft and the touchdown point. [1]

(iii) Calculate the horizontal ground distance BC from the point directly beneath the aircraft to touchdown, and evaluate $\sec^2\theta - \tan^2\theta$. [2]

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